

Pratham Aggarwal

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Education

University of California San Diego

Expected Mar 2027

Bachelor of Science in Data Science, Minor in Mathematics

GPA 3.6

Computer Science Courses: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Software Design, Artificial Intelligence, Databases (SQL/NoSQL), Machine Learning, Linear Algebra, Optimization

Honors: HDSI Scholarship Recipient, HKN-IEEE Award

Skills

Programming Languages	Python, SQL, C, C++, Java, JavaScript, TypeScript
DevOps & Tools	AWS, Google Cloud Platform (GCP), Linux, Docker, Git/GitHub, Shell Scripting
Software Development	React, Angular, Node.js, Flask, REST APIs, GraphQL, MongoDB, Redis
AI / Machine Learning	PowerBI, Tableau, Matplotlib, Pandas, PyTorch, TensorFlow, Scikit-Learn, OpenCV, CUDA, SciPy

Experience

Software Engineer @ Scripps Institution of Oceanography

Jun 2025 – Present

- Deployed a **React** web application on **Google Cloud** to evaluate **500+** climate model outputs, enabling interactive exploration of large datasets, earning a **\$4,500** scholarship for software development.
- Developed an ML benchmarking system with **20+** custom metrics and interactive time series visualizations, helping researchers compare models and identify accuracy gaps.
- Current:** Adapting **diffusion models** (EDM, DDPM) on sea surface temperature and solar irradiance data to predict 3D atmospheric fields, contributing to the AIMIP **open-source** initiative [Progress].

AI/ML Software Developer Intern @ SFIC Quantitative Technologies

Mar 2025 – Jun 2025

- Improved options trading returns by **12%** for a **\$1.3M** investment fund by developing a Deep Reinforcement Learning agent with adaptive policy learning and early exercise strategies.
- Improved trading decisions by **18%** through a sequential decision-making step that dynamically selected option type, expiration, and exercise timing from historical time-series data.
- Reduced stock price prediction error to **0.96 MSE** by training an **LSTM** model on **5 years** of market data with hyperparameter tuning and feature engineering.

Data Science Consultant Intern @ Solana Center

Mar 2025 – Jun 2025

- Led a **5-member** data analytics team to process **2,000+** composting records using **Python** and **SQL**, developing a real-time **Power BI** dashboard that improved evaluation accuracy by **6.7%** and guided program decisions.
- Boosted dataset reliability by **9%** through automated data-cleaning, validation, and imputation routines, ensuring consistent and reproducible analytics across stakeholders.
- Managed team workflows and version control using **Linux** and **Git**, improving collaboration and reporting efficiency.

Web Developer @ Existential Robotics Lab

Aug 2025 – Present

- Refined PyBullet simulation modules for mapping and localization, improving reproducibility by **40%** across **100+** trials.
- Benchmarked motion-planning algorithms (A*, RRT, MPC) in dynamic environments, reducing path-planning failure cases by **28%** and lowering average trajectory error below **0.15m**.
- Improved simulation speed by **2.1x** through parallel physics evaluations, allowing concurrent testing of **10+** algorithms.

Projects

QuantFlow - Fullstack Risk Management App | *Node.js, Flask, Random Forests, Docker, CI/CD*

Aug 2025

- Developed a risk platform using machine learning and Monte Carlo simulations, achieving **85%** accuracy in volatility forecasting and delivering real-time market data through **25+** API endpoints.
- Deployed containerized microservices using **Docker** containers making the app scale easily across Vercel and Render with automatic updates through **CI/CD** pipelines.

LLM-Based Conversational AI System | *Hugging Face, PyTorch, Semantic Embedding, NLP*

Sep 2025

- Built a conversational AI platform simulating **35k+** dialogues using Hugging Face LLMs (DialoGPT, MiniLM), providing insights into agent performance, response patterns, and failure cases.
- Developed an NLP telemetry pipeline processing **300k+** utterances from synthetic and real datasets, tracking intents, domains, and entities to improve retrieval accuracy by **25%** through semantic embeddings.

Hackathon Winner - Homeless Services Computer Vision Platform | *Scikit-learn, Pandas, OpenCV*

Jun 2025

- Built a geospatial forecasting platform that integrated **35+** transit, demographic, and geographic features, achieving **67%** accuracy in optimizing placement of homeless service centers.
- Deployed a real-time computer vision system with Oxen.ai and EyePop.ai to track transient populations, delivering actionable insights for data-driven service allocation.

ML-Based Energy Load Prediction | *Statistical Modeling, Hypothesis Testing, Matplotlib, Seaborn, EDA*

Jan 2025

- Achieved **91%** in predicting heating and cooling loads by training a multiple linear regression model on **750+** architectural and environmental samples.
- Improved forecasting accuracy by **12.3%** through k-means clustering to identify distinct consumption patterns, reducing operational costs by **9%**.